On the relation $\mathfrak{c} \to (\beta, n)^3_2$

Abstract

We investigate $\mathfrak{c} \to (\beta, n)^3_2$ for countable β and finite n. We prove the relation for all $\beta \leq \omega+1$ (every finite n) and for all countable β when $n \leq 3$. For $n \geq 4$ and $\beta > \omega + 1$, the natural transfinite-induction strategy requires a successor extension lemma that is **false**: we exhibit a coloring of $[\mathfrak{c}]^3$ with no 1-homogeneous 4-set in which a specific 0-homogeneous set of type ω admits no extension point. The counterexample does not refute the partition relation itself, only this proof strategy.

1 Notation

We identify each cardinal with the corresponding initial ordinal. Thus $\mathfrak{c}$ denotes the initial ordinal of cardinality $2^{\aleph_0}$. We use f for colorings.

For ordinals κ, α_0, α_1 and finite $r \geq 1$, the notation $\kappa \to (\alpha_0, \alpha_1)^r_2$ means: for every $f : [\kappa]^r \to \{0, 1\}$, either there exists $H \subseteq \kappa$ with $\mathrm{otp}(H) = \alpha_0$ and $f \restriction_{[H]^r} \equiv 0$, or there exists $H \subseteq \kappa$ with $\mathrm{otp}(H) = \alpha_1$ and $f \restriction_{[H]^r} \equiv 1$. The symmetric form $\kappa \to (\gamma)^r_2$ means every such f admits a monochromatic subset of order type γ.

2 Assumed external results

Ramsey's theorem. $\omega \to (\omega)^r_k$ for all finite $r \geq 1$ and $k \geq 2$.

Stepping-up lemma (Erdős–Rado). If κ is an infinite cardinal with $\kappa^{<\kappa} = \kappa$ and $\kappa \to (\alpha)^r_k$, then $2^\kappa \to (\alpha + 1)^{r+1}_k$.

3 Elementary reductions

Lemma 1 (Symmetric to asymmetric)**.** *If $\mu \to (\alpha)^3_2$, $\beta \leq \alpha$, and $n \leq \alpha$, then $\mu \to (\beta, n)^3_2$.*

Proof. Let $f : [\mu]^3 \to \{0, 1\}$. By hypothesis there is $H \subseteq \mu$ with $\mathrm{otp}(H) = \alpha$ and f constant on $[H]^3$. Fix the order-isomorphism $e : \alpha \to H$.

If $f \restriction_{[H]^3} \equiv 0$: the image $e[\beta]$ has order type β and $[e[\beta]]^3 \subseteq [H]^3$, so every triple from $e[\beta]$ has color 0.

If $f \restriction_{[H]^3} \equiv 1$: the image $e[n]$ has order type (equivalently, size) n and $[e[n]]^3 \subseteq [H]^3$. For $n < 3$, $[e[n]]^3 = \varnothing$, so the conclusion holds vacuously. □

Lemma 2 (Monotonicity in the domain)**.** *If $\mu \to (\beta, n)^3_2$ and $\nu \geq \mu$, then $\nu \to (\beta, n)^3_2$.*

Proof. Restrict any $f : [\nu]^3 \to \{0, 1\}$ to $[\mu]^3$. A witness in μ is also a witness in ν. □

4 Trivial second coordinate

Proposition 3. *Let $\beta \leq \kappa$.*

1. $\kappa \to (\beta, 2)^3_2$.
2. $\kappa \to (\beta, 3)^3_2$.

Proof. (1) Every 2-element set is vacuously 1-homogeneous for triples.

(2) If any triple has color 1, it is a 1-homogeneous 3-set. Otherwise every triple has color 0, and any $B \subseteq \kappa$ with $\mathrm{otp}(B) = \beta$ is 0-homogeneous. □

5 The first nontrivial case: $\beta \leq \omega + 1$

Theorem 4. $\mathfrak{c} \to (\omega + 1)^3_2$.

Proof. $\omega^{<\omega} = \omega$. Ramsey gives $\omega \to (\omega)^2_2$. The stepping-up lemma with $\kappa = \omega$, $\alpha = \omega$, $r = 2$, $k = 2$ yields $2^\omega \to (\omega + 1)^3_2$, i.e. $\mathfrak{c} \to (\omega + 1)^3_2$. □

Corollary 5. $\mathfrak{c} \to (\beta, n)^3_2$ *for every* $\beta \leq \omega + 1$ *and every* $2 \leq n < \omega$.

Proof. Set $\alpha = \omega+1$ in Lemma 1. Then $\beta \leq \alpha$ and $n \leq \alpha$ (since n is finite and $\omega+1$ is infinite). □

Note. The stepping-up lemma at $\kappa = \omega$ cannot yield $\mathfrak{c} \to (\omega+2)^3_2$, because the input $\omega \to (\omega+1)^2_2$ is false (every subset of ω has order type $\leq \omega$). The Baumgartner–Hajnal theorem gives $\omega_1 \to (\alpha)^2_k$ for all countable α, but stepping up from $\kappa = \omega_1$ produces the relation at $2^{\omega_1} > \mathfrak{c}$. A different approach is needed.

6 Ramsey reduction under the no-clique assumption

Lemma 6. *Let $f : [\mathfrak{c}]^3 \to \{0, 1\}$ and suppose there is no 1-homogeneous set of size n. Then every subset $S \subseteq \mathfrak{c}$ of order type ω contains a 0-homogeneous subset of order type ω.*

Proof. Let $e : \omega \to S$ be the increasing bijection. Define $g : [\omega]^3 \to \{0, 1\}$ by $g(\{i, j, k\}) = f(\{e(i), e(j), e(k)\})$. By Ramsey ($\omega \to (\omega)^3_2$), there is an infinite $I \subseteq \omega$ with g constant on $[I]^3$.

The image $e[I] \subseteq S$ has order type ω. If the constant color is 1, then $e[I]$ is an infinite 1-homogeneous set, and any n-element subset is a 1-homogeneous n-set, contradicting the assumption. So the constant color is 0. □

7 The successor extension lemma is false

The natural strategy to prove $\mathfrak{c} \to (\beta, n)^3_2$ for all countable β is transfinite induction: given a 0-homogeneous set A of order type γ, find a point $\xi > \sup A$ such that $A \cup \{\xi\}$ is 0-homogeneous of type $\gamma + 1$. This requires the following lemma.

> **Strong extension lemma (FALSE).** Let $f : [\mathfrak{c}]^3 \to \{0, 1\}$ with no 1-homogeneous set of size n. For every 0-homogeneous set $A \subseteq \mathfrak{c}$ and every $\delta > \sup A$, there exists $\xi > \delta$ such that $f(\{x, y, \xi\}) = 0$ for all $x < y$ in A.

Counterexample 7. There exists a coloring $f : [\mathfrak{c}]^3 \to \{0,1\}$ with no 1-homogeneous set of size 4 such that the set $A = \omega$ is 0-homogeneous but admits no extension point: for every $\xi \geq \omega$, $f(\{0,1,\xi\}) = 1$.

Proof. For increasing triples $x < y < z$ in $\mathfrak{c}$, define

$$f(\{x,y,z\}) = \begin{cases} 0 & \text{if } z < \omega, \\ 0 & \text{if } x \geq \omega, \\ 1 & \text{if } x < y < \omega \leq z \text{ and } (x,y) = (0,1), \\ 0 & \text{if } x < y < \omega \leq z \text{ and } (x,y) \neq (0,1), \\ 1 & \text{if } x < \omega \leq y < z \text{ and } x = 0, \\ 0 & \text{if } x < \omega \leq y < z \text{ and } x \neq 0. \end{cases}$$

$A = \omega$ is 0-homogeneous. Every triple $\{x,y,z\}$ with $x < y < z < \omega$ falls into the first case: color 0. ✓

No extension point. For any $\xi \geq \omega$, the triple $\{0,1,\xi\}$ satisfies $0 < 1 < \omega \leq \xi$ and $(x,y) = (0,1)$, so $f(\{0,1,\xi\}) = 1$. ✓

No 1-homogeneous 4-set. Let $F \subseteq \mathfrak{c}$ with $|F| = 4$.

Case $|F \cap \omega| = 4$. $F \subseteq \omega$: every triple has color 0 by the first clause.

Case $|F \cap \omega| = 0$. $F \subseteq [\omega, \mathfrak{c})$: every triple has color 0 by the second clause.

Case $|F \cap \omega| = 3$. F contains a triple entirely within ω: color 0.

Case $|F \cap \omega| = 1$, $F \cap \omega = \{a\}$. If $a \neq 0$: every triple involving a and two points $\geq \omega$ has color 0 (sixth clause); the triple of the three points $\geq \omega$ has color 0 (second clause). Not 1-homogeneous. If $a = 0$: the triples $\{0, \eta_i, \eta_j\}$ have color 1 (fifth clause), but $\{\eta_1, \eta_2, \eta_3\}$ has color 0 (second clause). Not 1-homogeneous.

Case $|F \cap \omega| = 2$, $F \cap \omega = \{a,b\}$, $a < b$. If $(a,b) \neq (0,1)$: the triples $\{a,b,\xi\}$ have color 0 (fourth clause). Not 1-homogeneous. If $(a,b) = (0,1)$: $\{0,1,\xi\}$ and $\{0,1,\eta\}$ have color 1; $\{0,\xi,\eta\}$ has color 1 (fifth clause). But $\{1,\xi,\eta\}$ has color 0 (sixth clause, $x = 1 \neq 0$). Not 1-homogeneous.

All cases exhausted. A 1-homogeneous set of size $m \geq 4$ would contain a 1-homogeneous 4-subset; since none exists, f has no 1-homogeneous set of any finite size ≥ 4. □

Remark 8. Counterexample 7 does **not** refute the partition relation $\mathfrak{c} \to (\beta, n)^3_2$. In the same coloring, the set $\{2,3,4,\ldots\} \cup \{\omega, \omega{+}1, \omega{+}2, \ldots\}$ is 0-homogeneous of order type $\omega \cdot 2$: every mixed triple $\{a,b,\xi\}$ with $2 \leq a < b < \omega \leq \xi$ satisfies $(a,b) \neq (0,1)$ (fourth clause, color 0), and every triple $\{a,\xi,\eta\}$ with $2 \leq a < \omega \leq \xi < \eta$ satisfies $a \neq 0$ (sixth clause, color 0). What fails is the *strategy* of extending a fixed, previously chosen set.

8 Why the weak extension lemma does not help

One might weaken the extension lemma to an existential form:

> **Weak extension lemma.** For every $\delta < \mathfrak{c}$ there exist *some* 0-homogeneous set $A \subseteq \delta$ and some $\xi > \delta$ with $f(\{x,y,\xi\}) = 0$ for all $x < y$ in A.

Even if this holds, it does not drive the induction. A set H of order type γ has already been constructed. The weak lemma provides a set A and a point ξ, but A may have no relationship to H: in general $H \not\subseteq A$, so there is no reason for $H \cup \{\xi\}$ to be 0-homogeneous.

9 Pattern extraction: what Ramsey does give

Under the no-clique assumption, the following is the strongest statement we can extract from Ramsey's theorem at the boundary.

Proposition 9 (Pattern extraction)**.** *Let $f : [\mathfrak{c}]^3 \to \{0,1\}$ with no 1-homogeneous set of size $n \geq 4$. Let $A \subseteq \mathfrak{c}$ be a countably infinite 0-homogeneous set. If no $\xi > \sup A$ extends A, then there exist $u < v$ in A and a set $X \subseteq (\sup A, \mathfrak{c})$ with $|X| = \mathfrak{c}$ such that $f(\{u, v, \xi\}) = 1$ for every $\xi \in X$.*

Proof. Since no $\xi > \sup A$ extends A, for each $\xi \in (\sup A, \mathfrak{c})$ there is a pair $p_\xi = \{x_\xi, y_\xi\} \in [A]^2$ with $f(\{x_\xi, y_\xi, \xi\}) = 1$. The set $[A]^2$ is countable and $|(\sup A, \mathfrak{c})| = |\mathfrak{c}|$ is uncountable, so by pigeonhole there is a fixed pair $\{u, v\} \in [A]^2$ and a set X of cardinality $\mathfrak{c}$ with $f(\{u, v, \xi\}) = 1$ for all $\xi \in X$. $\square$

Remark 10. From Proposition 9 one can apply Ramsey to further partition the points of X according to their interaction with u and v, obtaining an infinite set $M \subseteq X$ with a constant pattern $\sigma \in \{0,1\}^2$ for $(f(\{u, \xi, \eta\}), f(\{v, \xi, \eta\}))$ across all $\xi < \eta$ in M. In the pattern $(1,1)$, any two points of M together with u, v form a 1-homogeneous 4-set, contradicting the hypothesis when $n = 4$. For $n \geq 5$, a 1-homogeneous subset of M of size $n - 2$ would also yield a contradiction, but Ramsey on the countable set M only guarantees monochromatic subsets of order type ω, not of prescribed countable order type $\beta > \omega$. This is exactly where the argument stops.

10 Summary

Range	**Status**	**Reference**
$n = 2$, all countable β	proved	Proposition 3(1)
$n = 3$, all countable β	proved	Proposition 3(2)
$\beta \leq \omega + 1$, all finite n	proved	Corollary 5
$n \geq 4$, $\beta > \omega + 1$	gap present	§§ 7–9

Question 11. Is $\mathfrak{c} \to (\omega{+}2, 4)^3_2$ provable from Ramsey's theorem and the Erdős–Rado stepping-up lemma? This is the first case not settled by the methods of this note.